

Yonghao Yang

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EDUCATION

Wuhan University, School of Mathematics and Statistics

Expected Jul 2027

Major: Data Science and Big Data Technology

Major GPA: 93.4/100 (3.91/4.0) **Major Rank:** 1/63

Selected Coursework: Mathematical Analysis, Advanced Algebra, Probability Theory, Statistics, Real Analysis, Optimization, Numerical Analysis, Data Structures, Machine Learning, Deep Learning.

RESEARCH EXPERIENCE

Neural Models for NLP

Feb 2026 – Apr 2026

- Sentiment-analysis pipelines across N-gram, RNN, LSTM, Transformer, BERT with ablation study.
- Proposed CMLG multi-feature fusion with attention + LLM voting; improves macro F1. [\[DOI\]](#) [\[code\]](#)

Spatial Multi-omics with Deep Learning

Oct 2025 – Jan 2026

- Benchmarked GNN, VAE, GAN on single-cell and bulk RNA-seq in a unified PyTorch pipeline.
- Explored self-supervised, contrastive, and distillation objectives for spatial multi-omics alignment.

OPEN-SOURCE PROJECTS

- **machine-learning-handbook** — Knowledge base of DL fundamentals and math prerequisites. [\[code\]](#)
- **math-advanced-learning** — Notes on Functional Analysis and related graduate courses. [\[code\]](#)
- **basic-algorithms** — C++ templates for DP, graph, string, and number-theory algorithms. [\[code\]](#)

AWARDS & COMPETITIONS

National Scholarship

Ministry of Education, 2025

First-Class University Scholarship

Wuhan University, 2025

Merit Student

Wuhan University, 2025

Lanqiao Cup (Software & IT)

Provincial First Prize; National Third Prize, 2025

TECHNICAL SKILLS

Programming

Python, C / C++, R

ML / DL

PyTorch, NumPy, Pandas, scikit-learn

Tools

Git, LaTeX, Markdown, MkDocs

English

CET-4, CET-6

RESEARCH INTERESTS

Modern mathematical tools for training and analyzing deep generative & reasoning models.